USED CAR MARKET SEGMENTATION USING UNSUPERVISED LEARNING

1. Executive Summary

This project applies unsupervised machine learning to 3,577 used-car records to discover groups of vehicles with similar usage, age and ownership characteristics. The notebook compares K-Means, Hierarchical Clustering and DBSCAN. Six input features are used: kilometres driven, fuel, seller type, transmission, owner category and car age. Selling price and price category are kept out of model training and used only after clustering to interpret the discovered segments.

Key findings

- Ten clusters were selected for K-Means. The silhouette score for K = 10 was 0.2820, the highest among the tested K values (2–10).
- K-Means produced ten clusters, assigned all 3,577 cars and achieved the highest Calinski-Harabasz score (804.2531).
- DBSCAN, with eps = 0.50 and min_samples = 15, found 13 clusters and marked 612 records (17.11%) as noise. It achieved the highest Silhouette Score (0.3050) and lowest Davies-Bouldin Index (1.0009) among the three final models.
- Hierarchical Clustering produced ten clusters with a Silhouette Score of 0.2272.
- The two-dimensional PCA projection explained 64.41% of the total variance.

2. Objective and Scope

The objective is to identify natural segments in the used-car market without using selling price or price category to form the clusters. The resulting groups are then interpreted using price and vehicle characteristics.

1. Explore the used-car dataset and inspect its quality.
2. Select relevant vehicle features and preprocess numerical and categorical data.
3. Determine a suitable number of K-Means clusters.
4. Train and compare K-Means, Hierarchical Clustering and DBSCAN.
5. Profile the resulting car segments and visualise them using PCA.
6. Save cluster assignments, evaluation tables, plots and trained models.

3. Dataset Description

Property	Value
Source file	car.xlsx (sheet: Raw_data)
Records	3,577 used-car records
Columns	10
Missing values	0 in all columns
Duplicate rows	0

Average selling price	473,912.54
Average kilometres driven	69,250.55 km
Average car age	13.04 years
Target/label	No target label; unsupervised learning

The dataset contains name, year, selling_price, km_driven, fuel, seller_type, transmission, owner, car age and price category.

Features Used for Clustering

Feature type	Features
Numerical (2)	km_driven, car age
Categorical (4)	fuel, seller_type, transmission, owner
Excluded from training	name, year, selling_price, price category

The year and car age columns are redundant in this dataset: year + car age equals 2026 for every record. Therefore, car age was retained and year was excluded. Car name was excluded because it is a high-cardinality identifier. Selling price and price category were reserved for post-clustering interpretation.

4. Methodology

4.1 Exploratory Data Analysis

The notebook generated histograms for selling price, kilometres driven and car age; a correlation heatmap; and category-distribution plots for fuel, seller type, transmission, owner and price category.

4.2 Data Preprocessing

- Numerical features: median imputation followed by StandardScaler standardisation.
- Categorical features: most-frequent imputation followed by OneHotEncoder with unknown categories ignored.
- A ColumnTransformer combined the numerical and categorical pipelines.
- The six original features were transformed into a matrix of 3,577 rows and 17 encoded features.

4.3 Selecting the Number of Clusters

K	Inertia	Silhouette Score
2	8,984.0139	0.2721
3	7,556.3869	0.2669
4	6,605.5420	0.2433
5	5,959.2224	0.2397
6	5,443.8995	0.2471
7	5,062.3236	0.2421
8	4,731.7411	0.2499
9	4,419.8074	0.2740

10	4,183.2928	0.2820

The silhouette score was highest at K = 10 among the tested values, so K = 10 was selected for K-Means and Hierarchical Clustering. The elbow curve showed a steady decrease in inertia.

4.4 Algorithms and Evaluation Metrics

Algorithm	Configuration
K-Means	K = 10, random_state = 42, n_init = 10
Hierarchical Clustering	Agglomerative clustering, K = 10
DBSCAN	Grid search over eps values 0.5–3.0 and min_samples values 5, 10 and 15; selected eps = 0.50, min_samples = 15

Evaluation used Silhouette Score (higher is better), Davies-Bouldin Index (lower is better), Calinski-Harabasz Score (higher is better) and noise percentage for DBSCAN. Silhouette calculations were sampled on up to 2,000 records for K-Means selection.

5. Results

5.1 Final Algorithm Comparison

Algorithm	Clusters	Noise %	Silhouette	Davies-Bouldin	Calinski-Harabasz
DBSCAN	13	17.1093	0.3050	1.0009	447.5427
K-Means	10	0.0000	0.2820	1.3069	804.2531
Hierarchical	10	0.0000	0.2272	1.4969	581.7779

DBSCAN achieved the best Silhouette Score and Davies-Bouldin Index, while K-Means achieved the highest Calinski-Harabasz Score and assigned every record. Therefore, there is no single winner on every metric. DBSCAN gives stronger density-based separation among its assigned points, but leaves 17.11% of the dataset unassigned. K-Means is useful for a complete market segmentation and for producing the ten detailed car profiles below.

5.2 DBSCAN Hyperparameter Tuning

eps	min_samples	Clusters	Noise %	Silhouette
0.50	15	13	17.1093	0.3050
0.50	10	14	14.6212	0.2882
0.75	15	16	10.3998	0.2699
1.25	15	18	5.5633	0.2626
1.00	15	18	6.7375	0.2596
1.25	10	21	4.2494	0.2589
1.00	10	19	5.6472	0.2461

0.75	10	20	7.1289	0.2444

The selected setting was eps = 0.50 and min_samples = 15. It produced 13 clusters and 612 noise records. Configurations with eps values of 1.50 and above mostly collapsed into one cluster, so they did not provide a valid multi-cluster solution.

5.3 K-Means Cluster Profiles

Cluster	Cars	Avg age (yrs)	Avg km driven	Avg price	Common fuel	Seller	Transmission	Common owner	Price category
0	499	14.19	97872.95	402919.79	Diesel	Individual	Manual	Second Owner	Budget
1	538	8.98	28123.19	467092.89	Petrol	Individual	Manual	First Owner	Budget
2	304	14.63	67783.25	252373.93	Petrol	Individual	Manual	Second Owner	Budget
3	346	11.36	56478.0	838595.34	Diesel	Dealer	Manual	First Owner	Mid Range
4	477	9.52	42036.43	873379.42	Diesel	Individual	Manual	First Owner	Mid Range
5	279	10.7	36433.81	538261.59	Petrol	Dealer	Manual	First Owner	Budget
6	125	16.03	217970.56	364039.94	Diesel	Individual	Manual	First Owner	Budget
7	348	15.74	64727.04	207999.97	Petrol	Individual	Manual	First Owner	Budget
8	313	21.73	85342.44	116580.21	Petrol	Individual	Manual	Second Owner	Budget
9	348	13.03	106014.11	444830.44	Diesel	Individual	Manual	First Owner	Budget

Interpretation of the segments:

- Cluster 0: Older diesel cars with relatively high mileage, commonly second-owner and budget-category vehicles.
- Cluster 1: Comparatively younger, low-mileage petrol cars, commonly first-owner and budget-category vehicles.
- Cluster 2: Older petrol cars with moderate-to-high mileage, commonly second-owner and budget-category vehicles.
- Cluster 3: Diesel cars commonly sold by dealers, first-owner and mid-range by price, with an average selling price of about 838,595.
- Cluster 4: Relatively younger diesel cars with lower mileage and the highest average selling price among the ten clusters; commonly mid-range.

- Cluster 5: Petrol cars with lower average mileage, commonly dealer-sold, first-owner and budget-category.
- Cluster 6: Smallest segment (125 cars), with very high average mileage (about 217,971 km) and older diesel vehicles.
- Cluster 7: Older petrol cars, commonly first-owner and budget-category, with an average selling price of about 208,000.
- Cluster 8: Oldest segment (average 21.73 years), predominantly petrol, second-owner and budget-category; lowest average selling price (about 116,580).
- Cluster 9: Older diesel cars with high mileage, commonly first-owner and budget-category.

5.4 Price Category Cross-Check

Price category was not used to train the clustering models. After clustering, the notebook compared the price-category distribution within each K-Means cluster. Budget was the dominant category in clusters 0, 1, 2, 5, 6, 7, 8 and 9. Mid Range was the most common category in clusters 3 and 4. Cluster 8 was 99.0% Budget, while clusters 3 and 4 had larger Mid Range and Premium shares.

Cluster	Budget %	Mid Range %	Premium %
0	76.6	19.8	3.6
1	59.5	38.3	2.2
2	95.1	3.6	1.3
3	33.2	47.7	19.1
4	21.6	56.8	21.6
5	62.4	30.5	7.2
6	74.4	21.6	4.0
7	96.8	2.6	0.6
8	99.0	1.0	0.0
9	67.0	28.4	4.6

5.5 PCA Visualisation

Principal Component Analysis reduced the 17-dimensional preprocessed feature matrix to two dimensions for plotting. The first two principal components explain 64.41% of the total variance. The scatter plot provides a two-dimensional view of the ten K-Means segments, although some information is lost during dimensionality reduction.

6. Discussion and Limitations

- The dataset separates into ten K-Means groups based on a combination of vehicle age, kilometres driven, fuel, seller type, transmission and owner category.
- The silhouette values are moderate (0.2820 for K-Means and 0.2272 for Hierarchical), indicating that the segments are not perfectly separated.
- DBSCAN identifies denser local groups and outliers, but its 17.11% noise share means it does not assign every vehicle to a segment.

- Most cluster modes show Manual transmission, so transmission alone does not distinguish the segments strongly.
- The dataset contains no missing values or duplicate rows, but results are specific to the supplied dataset and feature choices.
- Price is reserved for post-clustering interpretation; therefore, the clusters should not be described as price-prediction outputs.

7. Conclusions and Recommendations

The project successfully grouped 3,577 used cars into ten interpretable K-Means segments. The segments reveal differences in vehicle age, usage, fuel type, seller type, ownership and post-clustering price patterns. DBSCAN provided the strongest Silhouette and Davies-Bouldin metrics among the compared algorithms, while K-Means provided complete coverage and the strongest Calinski-Harabasz score.

7. Use K-Means when the objective is to assign every listed car to one of ten market segments.
8. Use DBSCAN as a complementary method to identify dense groups and potentially unusual vehicles; review its 612 noise records separately.
9. Explore cluster stability across random seeds and alternative feature combinations.
10. Compare results using numerical features only and test other values of K to assess whether the ten-segment structure remains useful.
11. Validate the segments on a larger and more recent used-car dataset before applying them to market decisions.

8. Output Files and Execution Workflow

The notebook saves its generated artefacts in outputs_car_unsupervised_clustering.

Category	Generated outputs
Figures	feature_distributions.png, correlation_heatmap.png, fuel_distribution.png, seller_type_distribution.png, transmission_distribution.png, owner_distribution.png, price_category_distribution.png, elbow_method.png, silhouette_by_k.png, dbscan_neighbor_distances.png, cluster_sizes.png, pca_cluster_visualization.png
Tables / data	cluster_assignments.csv, clustering_model_comparison.csv, kmeans_k_selection.csv, cluster_profiles.csv, cluster_categories.csv, dbscan_tuning_results.csv, cluster_vs_price_category.csv
Models	preprocessor.joblib, k_means_model.joblib, hierarchical_clustering_model.joblib, dbscan_model.joblib
Metadata / report	model_metadata.json, final_project_report.txt

Notebook Execution Steps

12. Load car.xlsx from the Raw_data sheet and inspect shape, data types, missing values, duplicates and descriptive statistics.
13. Perform exploratory data analysis using distributions, category plots and a correlation heatmap.
14. Select six clustering features and preprocess numerical and categorical columns.
15. Run K-Means for K = 2 to 10 and compare inertia and silhouette scores.
16. Train K-Means and Hierarchical Clustering, then evaluate their cluster assignments.
17. Tune DBSCAN over eps and min_samples combinations and select the configured model.
18. Generate K-Means numerical and categorical cluster profiles and compare price categories after clustering.
19. Apply PCA for two-dimensional visualisation, compare algorithms and save the outputs and models.

Appendix: Important Execution Notes

The notebook expects a file named car.xlsx with a worksheet named Raw_data. The attached notebook contains the execution outputs used in this report. The source Excel dataset was not included among the supplied attachments, so the report documents the saved notebook results rather than independently rerunning the complete workflow.

All selling-price values are reported in the units present in the source dataset; the notebook does not explicitly document a currency symbol.